

Capacity Allocation Decision

Workload **wl-2026-0842** · facility iad-dc-01, Ashburn VA · conditions 'normal' · trace gm-17726d652539

APPROVED WITH CONDITIONS	Zone zone-c	Delay 14.0 h	Rounds 3	Capex \$48,000
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Plan

Deploy the 6-rack GB200 NVL72 workload in zone-c by utilizing 5 existing liquid-ready slots and retrofitting 1 additional rack for direct-to-chip liquid cooling, introducing a 14-hour installation delay and \$48,000.00 in one-time capex.

Conditions

- Retrofit 1 rack in zone-c for direct-to-chip liquid cooling
- Accept a 14-hour installation delay

How the conflicting verdicts were reconciled

The conflict was resolved by applying precedent prec-gm-0a313cb665da, which was originally unlocked by the Facilities agent's proposal to split/retrofit the deployment in zone-c. While Facilities and Cost initially favored zone-b to avoid capex, both Power and Cooling identified hard physical limits in zone-b (only 560.6 kW of electrical headroom and 2 CDU ports, whereas the workload requires 792.0 kW and 6 CDU ports). Zone-c is the only physically viable zone. By retrofitting only 1 rack in zone-c (utilizing 5 existing liquid-ready slots), we reduce the one-time capex to \$48,000.00 and enable a physically consistent deployment.

Trade-offs accepted

- Incurred a 14-hour delay and \$48,000.00 capex for retrofitting 1 rack in zone-c instead of deploying immediately with \$0.00 capex in zone-b
- Avoided severe power headroom deficit in zone-b (560.6 kW available vs 792.0 kW required)
- Avoided CDU port deficit in zone-b (2 available vs 6 required)
- Avoided floor load limit violation in zone-a (1200 kg limit vs 1360 kg rack weight)
- Gained a more efficient cooling environment in zone-c with a PUE of 1.226 compared to zone-b's PUE of 1.234

Cost and efficiency

Measure	Value	Reported by
Recurring monthly cost	\$1,370,015	cost agent
Change vs original request	+12.5%	cost agent
One-time capex	\$48,000	cost agent
Monthly saving vs next-best	not reported	cost agent
Payback period	not reported	cost agent
PUE at chosen zone	1.226	cooling agent
PUE at next-best zone	1.234	cooling agent

Zone-c is highly efficient with a PUE of 1.226, which is slightly better than the next-best viable liquid zone (zone-b) PUE of 1.234, though zone-b is physically ruled out due to power and cooling constraints.

Stranded capacity avoided

Avoided stranding power headroom in zone-a (which has 3887.2 kW headroom but is air-cooled and has a 1200 kg floor limit) and avoided stranding facilities capacity in zone-b (which has 8 liquid-ready racks but only 560.6 kW of power headroom and 2 CDU ports).

The negotiation, round by round

Round 1 is independent: no agent sees another's answer. Later rounds re-prompt each agent with the others' positions attached.

Round 1 — joint feasibility: **zone_mismatch**

Agent	Verdict	Zone	Reasoning
cooling	conditional	zone-c	The workload requires direct-to-chip liquid cooling at 132.0 kW per rack across 6 racks, totaling 792.0 kW IT power (approx 776.2 kW thermal load at 0.98 factor). Zone-c is the only viable liquid zone supporting the per-rack ceiling of 140 kW (above the 132 kW requirement) and having enough CDU ports (10 free ports vs 6 required). Zone-b has only 2 free CDU ports and is ruled out. Zone-a and zone-d are air-cooled and ruled out due to both the direct-to-chip requirement and a hard 40 kW/rack ceiling. However, Zone-c only has 4,129.9 kW of thermal headroom, and ambient temperature is 34.4 C with
cost	feasible	zone-b	Placing the 432 GB200 GPU workload (792 kW total across 6 racks) in zone-b is financially feasible. Zone-b is the preferred zone for direct-to-chip liquid cooling with existing DLC infrastructure (liquid_retrofit_cost_usd_per_rack is 0.0 USD). The recurring monthly cost for zone-b is \$1,210,334.48 (\$140,061.25 demand charge + \$1,070,267.63 energy, plus proportional capex). The next-best alternative zone-c requires a retrofit cost of \$48,000.00 per rack (\$288,000.00 total capex for 6 racks), resulting in a monthly cost of \$1,317,472.75. Choosing zone-b saves \$107,138.27 monthly compared to zone
facilities	conditional	zone-b	Workload requires 6 racks of GB200 NVL72 at 1360 kg each with direct-to-chip liquid cooling. Zone-b has 8 liquid-ready racks free with 1800 kg limit, which accommodates the weight and cooling. However, zone-b only has 8 available racks total, and zone-c has 5 liquid-ready racks and 3 retrofittable racks. Zone-c can satisfy the 6-rack requirement by retrofitting 1 rack at 14 hours delay. Zone-a is ruled out due to 1200 kg floor limit being lower than the 1360 kg rack weight. Zone-d is ruled out due to having 0 liquid-ready racks and requiring too many retrofits.
power	feasible	zone-a	Zone A has 3887.2 kW of headroom and 388 spare 30A/208V circuits, easily satisfying the 792.0 kW (6 racks at 132.0 kW) and approximately 14 circuits required per rack. Substation current load is 42.47 MW against 72.0 MW firm capacity (59.0%), and adding this load keeps utilization well below the 85.0% maximum sustained threshold. No planned outages affect SG-1 during the deployment window.

Round 2 — joint feasibility: **zone_mismatch**

Agent	Verdict	Zone	Reasoning
cooling	feasible	zone-c	The workload requires direct-to-chip liquid cooling at 132.0 kW per rack across 6 racks, totaling 776.16 kW of thermal load (0.98 x 792.0 kW). Zone-a and Zone-d are air-cooled CRAH zones with a hard 40.0 kW/rack ceiling, making them impossible for this 132.0 kW rack density. Power agent incorrectly endorsed Zone-a, which is ruled out by thermal per-rack limits. Zone-b has only 2 free CDU ports, which is insufficient for the 6 ports required. Zone-c is liquid-cooled with a max 140.0 kW/rack ceiling, has 10 free CDU ports (easily covering the 6 needed), and possesses 4129.9 kW of thermal headroom
cost	conditional	zone-c	Cooling and facilities agents have correctly established that the GB200 direct-to-chip liquid-cooled workload (792.0 kW across 6 racks) can only be hosted in zone-c, as zone-a and zone-d are air-cooled and zone-b lacks sufficient CDU ports. Zone-c requires 1 rack retrofit at \$48,000.00 capex plus 14 hours of installation labor at \$110.00/hr. Under Virginia Code Ann. Sec. 58.1-609.3(18), this qualifying data center equipment purchase is exempt from sales and use tax. The recurring monthly cost in zone-c is \$1,370,176.95 compared to \$547,433.88 in air-cooled zone-a (next-best zone by cost), yiel

Agent	Verdict	Zone	Reasoning
facilities	feasible	zone-b	Workload requires 6 racks of GB200 NVL72 at 1360.0 kg each, requiring direct-to-chip liquid cooling. Zone-b has 8 liquid-ready racks free with a floor load limit of 1800.0 kg/rack (exceeding the 1360.0 kg rack weight). Zone-b completely satisfies all physical and weight constraints without needing retrofits.
power	feasible	zone-c	The workload requires 6 racks of GB200 NVL72 at 132.0 kW per rack, totaling 792.0 kW. Zone-c has 3,053.0 kW of electrical headroom (breaker capacity 16,900 kW vs 13,847.0 kW allocated), 305 spare 30A/208V circuits (only ~84 needed for 792 kW based on 10 kW/circuit derated), and no planned outages on its upstream switchgear SG-2. Cooling and facilities peers endorse zone-c (with a minor 1-rack retrofit handled by facilities). Zone-b is ruled out electrically and thermally due to near-limit headroom (560.6 kW remaining, tight for expansion and prone to GS-5 penalties/overloads). Zone-a and zone-

Round 3 — joint feasibility: conditions_pending

Agent	Verdict	Zone	Reasoning
cooling	conditional	zone-c	The workload requires direct-to-chip liquid cooling for 6 racks at 132.0 kW per rack (totaling 792.0 kW thermal load, approx 776.2 kW IT-to-heat conversion). Air-cooled zones (zone-a and zone-d) have a hard ceiling of 40.0 kW/rack and are ruled out. Zone-b only has 2 free CDU ports, which is insufficient for the 6 ports required. Zone-c is a liquid-cooled zone with 10 free CDU ports, a max per-rack rating of 140.0 kW (accommodating the 132.0 kW request), thermal headroom of 4129.9 kW, and a current PUE of 1.226. Facilities and cost peers note zone-c requires 1 rack retrofit and 14 hours of del
cost	feasible	zone-c	Cooling and facilities agents have correctly ruled out zone-a, zone-b, and zone-d due to per-rack thermal limits, CDU port shortages, or cooling type incompatibilities. This leaves zone-c as the only physically viable zone capable of supporting 6 racks of GB200 NVL72 at 132 kW/rack with direct-to-chip cooling. Zone-c requires a liquid retrofit cost of \$48,000.0 per rack across 6 racks, totaling \$288,000.0 in one-time capex (install labor is \$110.0/hr). However, Virginia's sales-and-use tax exemption (Va. Code Ann. Sec. 58.1-609.3(18)) applies to qualifying data center equipment purchases, avoi
facilities	conditional	zone-c	The workload requires 6 racks of GB200 NVL72 at 1360.0 kg each, requiring direct-to-chip liquid cooling. Zone-a and zone-d are air-cooled and lack liquid readiness. Zone-b has 8 liquid-ready racks and 1800 kg floor limit, but zone-c provides 5 liquid-ready racks and 3 retrofittable racks (floor limit 1800 kg, exceeding the 1360 kg rack weight). To fulfill the 6-rack requirement in zone-c, 1 rack must be retrofitted for DLC, taking 14.0 hours using certified technicians.
power	feasible	zone-c	The workload requires 6 racks at 132.0 kW each, totaling 792.0 kW. Cooling and facilities agents have correctly ruled out zone-a, zone-b, and zone-d based on cooling and weight constraints. Zone-c has 3,053.0 kW of electrical headroom (breaker capacity 16,900.0 kW minus allocated 13,847.0 kW), 305 spare 30A/208V circuits (exceeding the ~84 circuits required for 792 kW under NEC continuous load derating), and a healthy 2N feed from SG-2 with no planned outages. Substation current load is 42.47 MW against a 72.0 MW firm capacity, so adding this load is well within limits.

Evidence each agent was shown

Captured by the harness at the moment of the decision, from the one database that agent is permitted to read. Compare these figures against the reasoning above.

cooling — read from cooling-db

zone	cooling type	current pue	free cdu ports	max kw per rack	thermal headroom kw
zone-a	air_crah	1.5	0	40.0	2,675.5
zone-b	liquid_dlc	1.2	2	140.0	1,013.4
zone-c	liquid_dlc	1.2	10	140.0	4,129.9
zone-d	air_crah	1.5	0	40.0	3,444.5

1 attempt(s) — accepted first pass, no guardrail correction

cost — read from cost-db

zone	cost per kw month usd	install labor rate usd per hour	liquid retrofit cost usd per rack
zone-a	12.4	95.0	0.0
zone-b	13.1	95.0	0.0
zone-c	12.9	110.0	48,000.0
zone-d	11.8	95.0	61,000.0

1 attempt(s) — accepted first pass, no guardrail correction

facilities — read from facilities-db

zone	available racks	floor load limit kg per rack	liquid ready racks	retrofitable racks
zone-a	30	1,200.0	0	0
zone-b	8	1,800.0	8	0
zone-c	8	1,800.0	5	3
zone-d	90	1,600.0	0	12

1 attempt(s) — accepted first pass, no guardrail correction

power — read from power-db

zone	allocated kw	breaker capacity kw	headroom kw	planned outage kw	spare 30a 208v circuits
zone-a	4,412.8	8,300.0	3,887.2	—	388
zone-b	12,639.4	13,200.0	560.6	—	56
zone-c	13,847.0	16,900.0	3,053.0	—	305
zone-d	1,995.4	7,000.0	5,004.6	yes	500

1 attempt(s) — accepted first pass, no guardrail correction

Provenance

The orchestrator holds no access to the power, cooling, facilities or cost databases — enforced by IAM Conditions, not convention. Every figure in this report was reported by a specialist agent and is attributable to its verdict above.

gcloud logging read 'jsonPayload.correlation_id="gm-17726d652539"'